

Medical Robotics

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Lecture 5

Parallel Kinematic Architectures: Definitions, Kinematics, Singularities, and a Worked Example

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1. Context and Motivation

This lecture introduces **parallel kinematic architectures** — a class of robot structures not typically covered in standard robotics courses, yet highly relevant in medical and surgical robotics. The overarching framework remains the reference design table introduced earlier in the course: for each surgical domain, we ask which kinematic architecture, control modality, and sensor/actuator set best satisfies the clinical requirements. In the previous lectures, serial and dedicated RCM architectures were treated; now we systematically address the parallel and hybrid families.

Parallel architectures differ from serial ones in a fundamental structural property: rather than forming a single open kinematic chain from base to end-effector, they form *closed kinematic loops*, with multiple independent chains connecting the fixed base to a common end-effector platform. This structural difference has sweeping consequences for workspace, accuracy, singularity behaviour, and the relative difficulty of the forward and inverse kinematic problems — consequences that are in many respects the exact *inverse* of those familiar from serial manipulators.

2. Serial vs. Parallel Kinematic Architectures

2.1. Properties of Serial Architectures

Serial robots consist of a single open chain of links and joints. Their key properties are:

Advantages: large reachable workspace; high dexterity.

Disadvantages:

- **Accumulated errors.** Flexibility at joints and in links propagates and adds up along the chain, reducing positional accuracy at the end-effector.
- **Low payload-to-weight ratio.** Each actuator must support the weight and inertia of all subsequent links and actuators. As a consequence, actuators at the base must be large and powerful, while those near the end-effector can be smaller. The payload-to-weight ratio of serial robots is inherently limited by the fact that proximal actuators must support all distal masses; typical industrial serial robots achieve ratios in the range of 0.1–0.5, substantially lower than parallel architectures where the load is shared across multiple chains.
- **High inertia.** The large number of moving parts leads to high inertia, requiring powerful motors and making the structure unsuitable for applications demanding high accelerations (flight simulators, fast pick-and-place tasks).
- **Inverse kinematics not always in closed form.** Depending on the architecture, closed-form inversion may not exist, requiring numerical methods with their associated convergence and reliability risks.

2.2. Properties of Parallel Architectures

In a parallel structure, multiple serial chains — each departing from the fixed base — are all connected to a common end-effector platform. The workspace of a parallel manipulator is essentially the *intersection* of the workspaces of its individual chains, which is why it is much smaller than the workspace of a comparable serial robot.

Advantages:

- **High rigidity and payload-to-weight ratio.** The load on the end-effector is shared among all the chains simultaneously; no single chain bears the full load. This allows a high

useful payload relative to the structure's own weight.

- **High accuracy.** Joint errors are also shared, and their effects at the end-effector partially cancel, rather than accumulating as in a serial chain.
- **Low inertia of moving parts, high agility.** Actuators can be located at the base, so only lightweight links need to move. This dramatically reduces the effective moving inertia, enabling high accelerations. It is for this reason that parallel architectures are the preferred choice for flight simulators and very fast industrial tasks.
- **Simpler inverse kinematics.** The inverse kinematics of parallel robots is generally simpler than their direct kinematics, and for many common architectures — including planar parallel manipulators and the Stewart platform — it decouples into independent per-chain subproblems that admit closed-form geometric solutions. However, this is not universally guaranteed: spatial parallel robots with complex chain geometries may still require numerical methods for inversion.

Disadvantages:

- **Limited workspace and low dexterity.** The intersection of multiple chain workspaces is considerably smaller than that of a single serial chain.
- **Complex direct kinematics.** The forward kinematic problem (finding the end-effector pose given the joint values) is generally not solvable in closed form and may have multiple solutions.
- **Internal singularities.** Unlike serial robots, parallel manipulators can encounter singularities *inside* the workspace, not only at its boundary.
- **Passive joints required.** To avoid over-constrained motion and the arising of internal forces, passive (unactuated) joints must be introduced in each chain.

Small workspace as an asset in medicine. In industrial robotics, a small workspace is clearly a drawback. In medical robotics, it can be a safety feature: a robot that can only reach a small volume is intrinsically limited in the harm it can cause in case of uncontrolled motion. This is one reason why parallel architectures appear in several patient-mounted or miniaturised surgical systems.

Is the inverse kinematics of parallel robots always in closed form? The claim that parallel manipulators *always* admit a closed-form inverse kinematic solution is a simplification that deserves qualification. What is more precisely true is that the inverse problem is generally *much simpler* than the direct problem, and for the majority of common architectures it does yield closed-form solutions — but this is not universally guaranteed.

The structural reason for the simplicity is the following. Once the desired pose of the end-effector platform is specified, the position of every platform attachment point B_i is immediately known. Each kinematic chain from base point A_i to B_i then becomes an *independent* serial manipulator with both endpoints fixed, whose inverse kinematics is an elementary subproblem. For short chains (one or two joints), this is always solvable in closed form — which covers the vast majority of practical parallel architectures, including the planar manipulators treated in these notes and the Stewart–Gough platform (for which the inverse kinematics reduces to computing $\rho_i = \|B_i - A_i\|$, a single square root per leg). For spatial parallel robots with longer or geometrically complex chains, closed-form solutions are not guaranteed.

Beyond the geometric method illustrated in the lecture, several systematic frameworks exist

for the inverse kinematics of closed-loop mechanisms:

- **Vector loop equations.** Closure conditions are written as vector equations for each chain and solved directly or after differentiation. This is the most general approach, applicable to any architecture without requiring geometric insight.
- **Screw theory / Plücker coordinates.** The kinematics is expressed using screw coordinates; the inverse problem often decomposes naturally along the screw axes of the joints.
- **Elimination theory and Gröbner bases.** When the inverse problem yields a polynomial system, systematic algebraic elimination (resultants, Gröbner bases) finds all solutions and characterises their number. This is the appropriate tool when closed-form solutions do not exist.

The general principle is therefore: *the inverse kinematics of a parallel robot decouples into independent per-chain subproblems that are usually, but not always, solvable in closed form — in contrast to the direct kinematics, for which closed-form solutions are the exception rather than the rule.*

2.3. Hybrid Serial/Parallel Architectures

When both a large workspace (or high dexterity) and high accuracy are required, the solution is a **hybrid architecture**: a parallel sub-structure for accurate, high-stiffness positioning of the wrist, followed by a serial sub-structure for orientation. The two sub-structures contribute complementary properties, at the cost of combining their respective disadvantages.

Representative examples mentioned in the lecture:

- **Surgiscope:** a surgical microscope in which a parallel base provides accurate spatial positioning while a serial wrist orients the microscope.
- **Robot for Ophthalmic Surgery:** developed at TU Munich, a microsurgery robot using parallelogram linkages to achieve high accuracy in a hybrid structure. The two prismatic joints at the base can move together (acting as a single prismatic) or differentially (creating a rotation via the parallelogram geometry), providing the necessary degrees of freedom with high stiffness. It uses virtual fixtures to guide the surgical gesture, as we will see in later lectures.
- **Parallel wrists (CERT/INRIA):** compact parallel wrist structures at the distal end of MIS instruments.

2.4. Examples of Parallel Architectures in Medical Robotics

Renaissance (spinal surgery). Renaissance is a 6-DOF Stewart platform, sterilisable, measuring only $50 \times 50 \times 80 \text{ mm}^3$ and weighing 250 g. Its workspace is $20 \times 20 \times 20 \text{ mm}^3$, and it positions the surgical drill guide with accuracy better than 0.1 mm. Because it is mounted directly on the patient's vertebra, it moves with the patient during any repositioning in the operating room, eliminating the need to re-register the surgical plan. The primary motivation for this design is accuracy, but the patient-mounted approach also addresses the registration problem directly.

XACT Robotics (interventional radiology). A parallel-architecture needle-insertion robot compatible with CT imaging. The parallel front-end positions and orients the needle guide according to the pre-operative plan; the needle is then inserted autonomously. This system was

developed to allow autonomous execution of percutaneous needle procedures under CT guidance, removing the surgeon from the radiation field. Unlike most needle-insertion systems where the surgeon performs the insertion manually, this system autonomously executes the insertion while tracking target motion due to tissue–needle interaction.

Miniaturised Stewart Platform (DVRK phantom). A low-cost (\$250) platform measuring $20 \times 20 \times 10$ cm, designed to fit inside the workspace of a da Vinci Research Kit surgical robot and simulate physiological organ motion (breathing, heartbeat, peristalsis) in phantom tissue experiments. Its translational range of motion is 1.27 cm along each axis.

3. Formal Definitions of Parallel Manipulators

Key definitions.

- **Generalised parallel manipulator:** a closed kinematic chain in which the end-effector is connected to the fixed base through several independent kinematic chains.
- **Parallel robot:** a generalised parallel manipulator whose end-effector has n degrees of freedom, formed by connecting the end-effector to the base via at least two independent chains, and equipped with exactly n actuators.
- **Completely (fully) parallel manipulator:** a parallel robot in which the number of kinematic chains equals the number of DOF of the end-effector, with each chain containing exactly one actuated joint.

Parallel robots may employ revolute (R), prismatic (P), universal (U, 2 DOF), or spherical (S, 3 DOF) joints. Both active (actuated, highlighted in red in the slides) and passive (unactuated) joints are present.

The Stewart–Gough platform is the canonical 6-DOF fully parallel manipulator: six legs, each with a prismatic actuated joint (the only active joint per leg) and revolute passive joints, connecting the base to the moving platform.

4. Kinematics of Planar Parallel Manipulators

4.1. Inverse Kinematics of the 3-RRR Planar Manipulator

The 3-RRR manipulator has three legs, each consisting of one *actuated* revolute joint (underlined) followed by two passive revolute joints. The three chains connect the fixed base points A_1, A_2, A_3 to the triangular end-effector platform at attachment points B_1, B_2, B_3 . The end-effector has 3 planar DOF: position (x_C, y_C) of its centroid C and orientation ϕ (angle of the body frame SR_1 relative to the world frame SR_0).

Problem: given (x_C, y_C, ϕ) , find the active joint angles $\theta_1, \theta_2, \theta_3$.

Method (geometric): for each leg i , the vector $A_i B_i$ can be expressed in two independent ways. The first uses the known geometry of the platform (the position and orientation of B_i in the world frame):

$$A_i B_i = A_i O + OC + {}^0 R_1 {}^1 C B_i \quad (1)$$

where ${}^1 C B_i$ are the coordinates of B_i in the body frame (known), and ${}^0 R_1$ is the rotation matrix from SR_1 to SR_0 (known from ϕ). The second uses the geometry of the i -th leg:

$$A_i B_i = A_i O + OM_i(\theta_i) + l_2^i \mathbf{n} \quad (2)$$

where M_i is the elbow joint position (a function of θ_i) and $\mathbf{n}$ is the unit vector along the second link. Equating (1) and (2):

$$A_iO + OC + {}^0R_1 {}^1CB_i = A_iO + OM_i(\theta_i) + l_2^i \mathbf{n} \quad (3)$$

Equation (3) involves only θ_i as the unknown (for each i separately). Geometrically, solving it corresponds to finding the elbow point M_i as the intersection of two circles: one of radius l_1^i centred at A_i , and one of radius l_2^i centred at B_i . Two circles intersect in at most 2 points, giving **2 solutions per leg** and **8 solutions total** for the manipulator (the elbow-up/elbow-down choice for each of the three legs independently).

Alternative interpretation. Each chain from A_i to B_i is itself a 2R planar serial manipulator with a *known* end-effector position (since both A_i and B_i are now known once the platform pose is given). The inverse kinematics of a 2R serial planar manipulator is elementary and always in closed form. This confirms the general principle: *the inverse kinematics of a parallel manipulator reduces to the inverse kinematics of its individual serial chains, which might be easily solvable geometrically.*

4.2. Direct Kinematics of the 3-RPR Planar Manipulator

The 3-RPR manipulator has three legs, each consisting of a passive revolute joint at the base, an *actuated* prismatic joint (leg length ρ_i), and a passive revolute joint at the platform. The end-effector is a triangular platform with reference point $B(x, y)$ and orientation φ .

Problem: given the three actuated joint lengths ρ_1, ρ_2, ρ_3 , find (x, y, φ) .

Writing the three distance equations from the geometry of each leg:

$$\rho_1^2 = x^2 + y^2, \quad \rho_2^2 = (x + l_2 \cos \varphi - x_2)^2 + (y + l_2 \sin \varphi - y_2)^2, \quad \rho_3^2 = (x + l_3 \cos(\varphi + \theta) - x_3)^2 + \dots \quad (4)$$

Subtracting the first equation from the second and third linearises the problem in x and y :

$$\rho_2^2 - \rho_1^2 = Rx + Sy + Q, \quad \rho_3^2 - \rho_1^2 = Ux + Vy + W$$

This linear system yields x and y as explicit functions of φ alone:

$$x = -\frac{SA_1 - VA_2}{\Delta}, \quad y = \frac{RA_1 - UA_2}{\Delta}$$

Substituting these expressions back into the first equation of (4) produces a single equation in φ . Using the substitution $T = \tan(\varphi/2)$, this becomes a **degree-6 polynomial** in T :

$$C_0 + C_1T + C_2T^2 + C_3T^3 + C_4T^4 + C_5T^5 + C_6T^6 = 0 \quad (5)$$

where the coefficients C_i depend only on the known geometry of the manipulator. A degree-6 polynomial has up to 6 roots, none of which can in general be expressed in closed form. This confirms that the **direct kinematics of parallel manipulators is usually not in closed form** even for a simple planar structure, in contrast to the situation for serial manipulators.

5. Singular Configurations of Parallel Manipulators

5.1. General Framework

The kinematic relationship between joint coordinates $\boldsymbol{\theta}$ and the end-effector generalised coordinates $\mathbf{X}$ of a parallel manipulator can always be written implicitly as:

$$\mathbf{H}(\mathbf{X}, \boldsymbol{\theta}) = \mathbf{0}$$

Differentiating with respect to time:

$$\mathbf{U}\dot{\mathbf{X}} + \mathbf{V}\dot{\boldsymbol{\theta}} = \mathbf{0} \quad (7)$$

where $\mathbf{U} = \partial\mathbf{H}/\partial\mathbf{X}$ and $\mathbf{V} = \partial\mathbf{H}/\partial\boldsymbol{\theta}$. This is the fundamental differential kinematic equation for parallel manipulators. Note that it contains *two* matrices that can each lose rank, whereas for a serial manipulator the analogous equation $\dot{\mathbf{X}} = \mathbf{J}\dot{\boldsymbol{\theta}}$ contains only one Jacobian $\mathbf{J}$.

Three types of singularity in parallel manipulators.

- **First kind (serial / type-1):** $\mathbf{V}$ loses rank but $\mathbf{U}$ does not. There exist joint velocities $\dot{\boldsymbol{\theta}} \neq \mathbf{0}$ that produce no end-effector motion ($\dot{\mathbf{X}} = \mathbf{0}$). These are the direct analogue of serial manipulator singularities (where the Jacobian $\mathbf{J}$ loses rank), and they typically occur at the *boundary* of the workspace.
- **Second kind (parallel / type-2 / local):** $\mathbf{U}$ loses rank while $\mathbf{V}$ is full rank. There exists an end-effector velocity $\dot{\mathbf{X}} \neq \mathbf{0}$ achievable with all actuators locked ($\dot{\boldsymbol{\theta}} = \mathbf{0}$). The mechanism gains an uncontrollable infinitesimal degree of freedom. These singularities can occur *inside* the workspace and are specific to parallel structures.
- **Third kind:** both $\mathbf{U}$ and $\mathbf{V}$ lose rank simultaneously. These are both configuration-dependent and architecture-dependent, occurring only under special geometric conditions.

The duality of force and velocity. The singularity conditions have direct static (force) interpretations due to the duality between the Jacobian and its transpose (which maps end-effector forces to joint torques). At a first-kind singularity, a force applied at the end-effector along the direction of the aligned links requires no torque from the actuators to be resisted — the structure is locked in that direction. At a second-kind singularity, the mechanism cannot resist a force applied at the end-effector in the direction corresponding to the uncontrollable motion, even with all actuators locked.

5.2. Singularities of the 3-RRR Planar Manipulator

For a symmetric 3-RRR with equilateral triangular base and platform (all base radii $a_1 = a_2 = a_3$ and all platform radii $b_1 = b_2 = b_3$):

5.2.1. First-kind (serial) singularities

One or more legs are fully extended or fully folded. A force applied at the end-effector along the direction of the collinear links of a leg produces no torque at the active joint of that leg. These configurations lie at the *boundary of the workspace* and can be avoided by confining the motion to the interior.

5.2.2. Second-kind (parallel) singularities

Two distinct cases arise, both involving end-effector motion with all active joints locked:

1. **Infinitesimal rotation** (first configuration in the corresponding slide) of the end-effector about its centroid is possible with all active joints locked. The mechanism cannot resist a torque applied at the end-effector.
2. **Infinitesimal translation** (second configuration in the corresponding slide) of the end-effector in the direction orthogonal to one of the second links is possible with all active joints locked. The mechanism cannot resist a force in that direction.

These singularities occur *inside the workspace*, which makes them particularly dangerous in medical applications.

5.2.3. Third-kind singularities

These occur under two distinct geometric conditions:

1. When the end-effector triangle has a *specific relationship* between its size and the link lengths: the end-effector can undergo arbitrary rotations about its centroid while the active joints are locked (second-kind component), and simultaneously the active joints can undergo arbitrary rotations while the end-effector does not move (first-kind component).
2. When the end-effector triangle has the *same size* as the base triangle and all links have equal length: the active joints can rotate arbitrarily without moving the end-effector.

6. Worked Example: RRRR Four-Bar Linkage with Endoscopic Camera (Homework 1 Solution)

This worked example, drawn directly from a past homework assignment, applies the general theory of parallel manipulator kinematics and singularities to a concrete closed-loop mechanism that has a direct connection to a real surgical robotic system.

6.1. Problem Statement

Consider a planar RRRR four-bar linkage in which all four links have length ℓ , a camera is mounted at the end of a rigid extension of length d attached to the third link, and the only actuated joint is J_1 at the base. The four questions are:

1. Determine the degrees of mobility of the camera.
2. Define the task describing the camera motion using a minimum number of parameters.
3. Determine the differential kinematics for the task defined in question 2.
4. Determine the singularities of the mechanism.

Connection to the Da Vinci surgical system. This planar four-bar linkage is kinematically equivalent to the parallelogram RCM mechanism used in the da Vinci patient-side manipulator. In the real system, the link $\overline{AD}$ (which would cross through the patient's body) is physically absent, and a double-parallelogram arrangement achieves the same kinematics without the offending link. In the plane of the parallelogram, with the penetration depth d fixed, the only remaining degree of freedom is the camera orientation. This homework therefore analyses the kinematic structure at the core of the world's most widely deployed surgical robotic system.

6.2. Question 1: Degrees of Mobility

The degrees of mobility m of a planar linkage are given by the Grübler formula for the plane:

$$m = \sum_{i=1}^n d_i - 3(n + 1 - l)$$

where n is the number of joints, d_i is the DOF of joint i , and l is the number of rigid links (including the base). For this mechanism: 4 revolute joints (each with $d_i = 1$), 4 moving links plus the base ($l = 4 + 1 = 5$, but since it is a closed loop $n = l$):

$$m = 4 \cdot 1 - 3(4 + 1 - 4) = 4 - 3 = 1$$

The camera has 1 degree of mobility: it can only be oriented, not translated (while the penetration depth d is fixed). This is precisely the RCM behaviour: the tool tip can rotate about the trocar insertion point, but the insertion point itself is the fixed centre of motion.

6.3. Question 2: Task Parameterisation

Since the camera has 1 DOF, its motion can be described by a single parameter. The most natural choice is the angle ϑ_c , defined as the orientation of the third link (the link carrying the camera). This angle is equal to the sum of the three joint angles:

$$\vartheta_c = \vartheta_1 + \vartheta_2 + \vartheta_3$$

Why not a Cartesian coordinate? One could alternatively use, say, the x -coordinate of the camera. However, the angular parameterisation is preferable because it leads directly to a simpler differential kinematic expression, as the subsequent derivation shows. The general principle is that the task variable should be chosen to match the geometric structure of the motion.

6.4. Question 3: Differential Kinematics

6.4.1. Setting up the closure constraint

Taking the time derivative of the task definition:

$$\dot{\vartheta}_c = \dot{\vartheta}_1 + \dot{\vartheta}_2 + \dot{\vartheta}_3$$

The passive joint velocities $\dot{\vartheta}_2$ and $\dot{\vartheta}_3$ must be related to the actuated joint velocity $\dot{\vartheta}_1$ through the closure constraint of the four-bar linkage.

To apply the closure constraint, the kinematic chain is *virtually opened* at point D (the location of joint J_4). This transforms the closed chain into an open 3R planar serial manipulator with joints J_1, J_2, J_3 and equal link lengths ℓ . In the open chain, the three joint angles are temporarily treated as independent; the point D is free to move.

The closure constraint is then *enforced at the velocity level*: the velocity of point D (as computed from the open chain) is set to zero. This means that the joint velocities must lie in the null space of the Jacobian of the open chain evaluated at D :

$$\begin{pmatrix} s_1 + s_{12} + s_{123} & s_{12} + s_{123} & s_{123} \\ c_1 + c_{12} + c_{123} & c_{12} + c_{123} & c_{123} \end{pmatrix} \begin{pmatrix} \dot{\vartheta}_1 \\ \dot{\vartheta}_2 \\ \dot{\vartheta}_3 \end{pmatrix} = \mathbf{0}$$

where $s_1 = \sin \vartheta_1$, $c_{12} = \cos(\vartheta_1 + \vartheta_2)$, etc.

6.4.2. Separating active and passive variables

Separating the actuated variable $\dot{\vartheta}_1$ from the passive variables $\dot{\vartheta}_2$ and $\dot{\vartheta}_3$:

$$\begin{pmatrix} s_1 + s_{12} + s_{123} \\ c_1 + c_{12} + c_{123} \end{pmatrix} \dot{\vartheta}_1 = - \begin{pmatrix} s_{12} + s_{123} & s_{123} \\ c_{12} + c_{123} & c_{123} \end{pmatrix} \begin{pmatrix} \dot{\vartheta}_2 \\ \dot{\vartheta}_3 \end{pmatrix}$$

Solving for the passive joint velocities:

$$\begin{pmatrix} \dot{\vartheta}_2 \\ \dot{\vartheta}_3 \end{pmatrix} = \frac{1}{s_3} \begin{pmatrix} -s_{23} - s_3 \\ s_{23} + s_2 \end{pmatrix} \dot{\vartheta}_1$$

6.4.3. Final differential kinematics

Substituting the passive joint velocities into $\dot{\vartheta}_c = \dot{\vartheta}_1 + \dot{\vartheta}_2 + \dot{\vartheta}_3$ and simplifying:

$$\dot{\vartheta}_c = \dot{\vartheta}_1 + \frac{1}{s_3} [(-s_{23} - s_3) + (s_{23} + s_2)] \dot{\vartheta}_1 = \dot{\vartheta}_1 \cdot \frac{s_2 - s_3 + s_3}{s_3} \cdot \frac{s_3}{s_3}$$

After simplification (all s_{23} terms cancel):

$$\boxed{\dot{\vartheta}_c = \frac{s_2}{s_3} \dot{\vartheta}_1}$$

This is the complete differential kinematic model of the camera orientation. It maps the actuated joint velocity $\dot{\vartheta}_1$ to the camera angular velocity $\dot{\vartheta}_c$ through the scalar Jacobian s_2/s_3 .

Written in the standard parallel manipulator form $U\dot{\vartheta}_c + V\dot{\vartheta}_1 = 0$:

$$s_3 \dot{\vartheta}_c = s_2 \dot{\vartheta}_1$$

where $U = s_3$ (scalar) and $V = -s_2$ (scalar).

6.5. Question 4: Singularity Analysis

From the differential kinematic equation $s_3 \dot{\vartheta}_c = s_2 \dot{\vartheta}_1$, singularities arise whenever one or both of the scalar quantities $s_2 = \sin \vartheta_2$ or $s_3 = \sin \vartheta_3$ vanish.

6.5.1. Serial singularities (first kind): $s_2 = 0$

$$\vartheta_2 = 0 \quad \text{or} \quad \vartheta_2 = \pm\pi$$

When $s_2 = 0$: the right-hand side of the kinematic equation vanishes, so $s_3 \dot{\vartheta}_c = 0$ regardless of $\dot{\vartheta}_1$. The actuated joint can move freely without producing any camera motion.

Physical interpretation: when $\vartheta_2 = \pm\pi$, the points A and C coincide ($A \equiv C$) and the point B can rotate around A under the action of J_1 without rotating the third link.

6.5.2. Parallel singularities (second kind): $s_3 = 0$

$$\vartheta_3 = 0 \quad \text{or} \quad \vartheta_3 = \pm\pi$$

When $s_3 = 0$: the left-hand side vanishes, so $\dot{\vartheta}_c$ can be non-zero even with $\dot{\vartheta}_1 = 0$. The camera can rotate with the actuated joint locked.

Physical interpretation: when $\vartheta_3 = \pm\pi$, the points B and D coincide ($B \equiv D$) and the point C (carrying the camera) can rotate freely around D without requiring any motion of J_1 . The passive joints J_2 and J_3 can accommodate this motion by themselves. The mechanism has gained an uncontrollable degree of freedom.

6.5.3. Third-kind singularities: $s_2 = 0$ and $s_3 = 0$

Both conditions must be satisfied simultaneously:

$$\vartheta_2 \in \{0, \pm\pi\} \quad \text{and} \quad \vartheta_3 \in \{0, \pm\pi\}$$

For the four-bar linkage with all links of equal length, s_2 and s_3 must simultaneously vanish. Both conditions can only be satisfied simultaneously at *degenerate configurations* where multiple links are collinear. These are isolated points in the configuration space.

6.5.4. Geometric representation in the $(\vartheta_2, \vartheta_3)$ configuration space

The singularity locus can be visualised on the torus $[-\pi, \pi] \times [-\pi, \pi]$ (the correct topology of the configuration space for two revolute passive joints): see last slide of the file HM1_Solution.pdf.

Practical consequence for the Da Vinci. The parallel singularity at $\vartheta_3 = \pm\pi$ (links BC and CD collinear, i.e., $B \equiv D$) corresponds to the parallelogram collapsing — a configuration in which the RCM constraint is lost and the tool tip can move uncontrollably. This degenerate configuration is prevented in the real da Vinci system through mechanical joint limits and software limits that keep the mechanism well away from this singularity during operation.

7. Summary and Key Conceptual Threads

- **Parallel and serial manipulators have complementary kinematic properties.** Serial: large workspace, complex inverse kinematics (sometimes); parallel: small workspace, simple inverse kinematics, complex direct kinematics. Hybrid structures exploit both.
- **The inverse kinematics of a parallel manipulator is often geometrically solvable** by reducing each chain to an independent serial subproblem with known end-points. Multiple solutions exist (e.g., 2 per chain, 2^n total for n chains).
- **The direct kinematics of a parallel manipulator is generally not in closed form.** For the 3-RPR, it reduces to a degree-6 polynomial. This is the price paid for the simplicity of the inverse problem.
- **Parallel manipulators have three types of singularity**, of which the second kind (parallel singularities, where the end-effector can move with actuators locked) is specific to closed-loop structures and can occur *inside* the workspace. This has direct safety implications in medical applications.
- **A popular method for solving the differential kinematics of closed-loop chains** is: (1) open the chain at a suitable point; (2) write the Jacobian of the resulting open chain; (3) enforce the closure constraint at velocity level by setting the velocity of the cut point to zero; (4) separate active from passive joint velocities; (5) substitute the passive velocities back into the task kinematics.
- **The four-bar linkage with an endoscopic camera** is the kinematic equivalent of the da Vinci parallelogram RCM mechanism. Its differential kinematics $s_3\dot{\vartheta}_c = s_2\dot{\vartheta}_1$ reveals serial singularities when $s_2 = 0$ and parallel singularities when $s_3 = 0$, with a clear physical interpretation for each.
- **Small workspace as a medical safety feature.** The limited workspace of parallel manipulators is not only acceptable but desirable in patient-mounted or miniaturised surgical systems where the reachable volume bounds the possible harm.

Key topics for revision.

1. Serial vs. parallel architecture trade-offs: workspace, accuracy, inertia, kinematic inversion, singularity structure.
2. Formal definitions: generalised parallel manipulator, parallel robot, completely parallel manipulator.
3. Inverse kinematics of the 3-RRR: the geometric construction, the two-circles intersection, number of solutions.

4. Direct kinematics of the 3-RPR: why it leads to a degree-6 polynomial, and what this implies for computational tractability.
5. The general differential kinematic equation $\mathbf{U}\dot{\mathbf{X}} + \mathbf{V}\dot{\boldsymbol{\theta}} = \mathbf{0}$ and the three types of singularity.
6. Singularities of the 3-RRR: physical interpretation of each kind.
7. Homework worked example: Grübler formula application, task parameterisation, virtual chain opening, enforcement of the closure constraint, separation of active/passive variables, singularity identification.
8. The connection between the four-bar linkage and the da Vinci parallelogram RCM mechanism.

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